HW 2 — Fresnel Equations (s pol.) & Brewster's Angle

How to read this: black = write this down · red = the answer · teal = explanation, no need to copy · grey box = optional · tags like (Lec 3, slide 9) give the source slide.

Problem 1 — Fresnel equations for s polarization

A plane wave goes from medium (μ, ε) into medium (μ′, ε′) across the plane z = 0, at angle of incidence i. The electric field is **perpendicular to the plane of incidence** (s polarization). Find the reflected and transmitted amplitudes — i.e. derive Jackson Eq. (7.39).

ANSWER

Impedance form (Lecture 4), $Z = \sqrt{\mu/\varepsilon}$:

$$\frac{E_0'}{E_0} = \frac{2Z'\cos i}{Z'\cos i + Z\cos r}, \qquad \frac{E_0''}{E_0} = \frac{Z'\cos i - Z\cos r}{Z'\cos i + Z\cos r}$$

The same result in n, n', i — **this is Eq. (7.39)**, the form asked for:

$$\frac{E_0'}{E_0} = \frac{2n\cos i}{n\cos i + \frac{\mu}{\mu'}\sqrt{n'^2 - n^2\sin^2 i}}$$

$$\frac{E_0''}{E_0} = \frac{n\cos i - \frac{\mu}{\mu'}\sqrt{n'^2 - n^2\sin^2 i}}{n\cos i + \frac{\mu}{\mu'}\sqrt{n'^2 - n^2\sin^2 i}}$$

Step 0 — How the professor sets up any interface problem

Lecture 3, slides 6–12 — learn this recipe once; every reflection problem follows it.

1. Write three waves and list what is unknown

incident (known): $\vec{E} = \vec{E}_0 e^{i(\vec{k}\cdot\vec{x} - \omega t)}$, $\vec{B} = \sqrt{\mu\varepsilon}\,\frac{\vec{k}\times\vec{E}}{k}$ (Lec 3, slide 8)

reflected: $\vec{E}'' = \vec{E}_0'' e^{i(\vec{k}''\cdot\vec{x} - \omega'' t)}$ transmitted: $\vec{E}' = \vec{E}_0' e^{i(\vec{k}'\cdot\vec{x} - \omega' t)}$ (Lec 3, slides 8–9)

unknown: $\vec{E}_0'$, $\vec{E}_0''$, $\vec{k}'$, $\vec{k}''$, ω', ω'' — the professor's own list, slide 9

2. Kinematics first — this kills the direction unknowns

The boundary conditions hold at every point of $z = 0$ and every time — only possible if all three phases are equal there.

$\omega = \omega' = \omega''$ and $\vec{k}\cdot\vec{x} = \vec{k}'\cdot\vec{x} = \vec{k}''\cdot\vec{x}$ Lec 3, slide 11

$\Rightarrow\ k\sin i = k''\sin r' = k'\sin r \ \Rightarrow\ r' = i$ and $n\sin i = n'\sin r$ Lec 3, slide 12

3. Dynamics second — only the amplitudes are left

Two unknowns remain, $\vec{E}_0'$ and $\vec{E}_0''$, so we need two equations. They come from Maxwell at the surface:

① $[\ \varepsilon(\vec{E}_0+\vec{E}_0'') - \varepsilon'\vec{E}_0'\ ]\cdot\hat{n} = 0$ — $D_\perp$ continuous

② $(\ \vec{E}_0+\vec{E}_0'' - \vec{E}_0'\ )\times\hat{n} = 0$ — $E_\parallel$ continuous

③ $[\ \vec{k}\times\vec{E}_0 + \vec{k}''\times\vec{E}_0'' - \vec{k}'\times\vec{E}_0'\ ]\cdot\hat{n} = 0$ — $B_\perp$ continuous

④ $\left[\ \frac{1}{\mu}(\vec{k}\times\vec{E}_0 + \vec{k}''\times\vec{E}_0'') - \frac{1}{\mu'}(\vec{k}'\times\vec{E}_0')\ \right]\times\hat{n} = 0$ Lec 3, slide 10 = Jackson 7.37 — $H_\parallel$ continuous

4. Choose a polarization

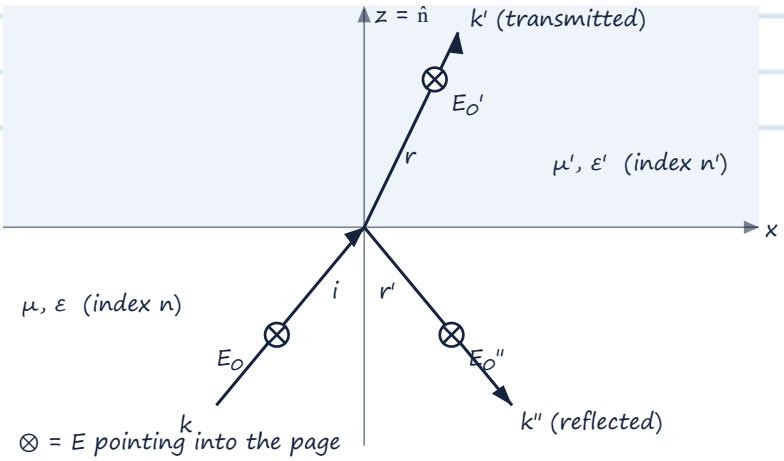
E *can point anywhere perpendicular to* ***k****. Split into the two natural cases:* ***E*** *in the plane of incidence (p) — done in class, Lecture 4 slides 2–3 — and* ***E*** *perpendicular to it (s), which is this homework (Lecture 4 slide 4, marked "HW 2").*

STARTING POINT — this is all you carry into Step 1:

- the four boundary conditions ①–④ above;
- $r' = i$ and $n\sin i = n'\sin r$;
- s polarization ⇒ **E** ⊥ plane of incidence.

Step 1 — Set up coordinates and write the vectors

Same axes as the professor's picture (Lecture 3, slide 7): interface at $z = 0$ so $\hat{n} = \hat{z}$, plane of incidence = x–z. Then "E perpendicular to the plane of incidence" means **E along $\hat{y}$.**



Angles named as the professor does: i incident, r' reflected, r transmitted. Step 0 already showed $r' = i$.

$\vec{E}_0 = E_0\hat{y}$, $\vec{E}_0'' = E_0''\hat{y}$, $\vec{E}_0' = E_0'\hat{y}$ — the two unknowns are E_0', E_0''

$\vec{k} = k(\sin i\,\hat{x} + \cos i\,\hat{z})$, $\vec{k}' = k'(\sin r\,\hat{x} + \cos r\,\hat{z})$

$\vec{k}'' = k(\sin r'\,\hat{x} - \cos r'\,\hat{z}) = k(\sin i\,\hat{x} - \cos i\,\hat{z})$ — using $r' = i$ from Step 0

Where these three lines come from

- **E along $\hat{y}$** — the plane of incidence is x–z, so "perpendicular to it" can only mean $\pm\hat{y}$. Allowed for all three waves because **$\vec{k}\cdot\vec{E} = 0$** and all three **$\vec{k}$** lie in the x–z plane.
- **$\vec{k}$ and $\vec{k}'$** — a direction at angle θ measured from the normal $\hat{z}$ is $\sin\theta\hat{x} + \cos\theta\hat{z}$. Multiply by the magnitudes $|\vec{k}| = \omega\sqrt{(\mu\varepsilon)}$, $|\vec{k}'| = \omega\sqrt{(\mu'\varepsilon')}$. Lec 3, slide 9
- **$\vec{k}''$** — minus on $\hat{z}$ because the reflected wave goes back down; magnitude is **k**, not k'', since $|\vec{k}''| = \omega''\sqrt{(\mu\varepsilon)} = \omega\sqrt{(\mu\varepsilon)} = k$. Lec 3, slide 9

Step 2 — Which of the four boundary conditions are useful?

The professor ticks the two that work and moves on (Lecture 4, slide 2). For s polarization they turn out to be the same two he used for p: ② and ④.

> **Reminder — the four conditions from p. 2** (Lec 3, slide 10):
> ① $D_\perp$ continuous (normal **D**) ② $E_\parallel$ continuous (tangential **E**)
> ③ $B_\perp$ continuous (normal **B**) ④ $H_\parallel$ continuous (tangential **H**)

① $D_\perp$: **E** is along $\hat{y}$, which is tangential, so $D_\perp = 0$ on both sides ⇒ $0 = 0$, nothing.

③ $B_\perp$: gives $k \sin i\,(E_0 + E_0'') = k' \sin r\, E_0'$, and Snell ($k \sin i = k' \sin r$) lets you divide that common factor off both sides, leaving $E_0 + E_0'' = E_0'$ — the **same equation as ②** ⇒ nothing new.

✓ ② and ✓ ④ are the two that give the answer.

Step 3 — Write out ② and ④

Read the tangential components straight off the figure, using $|H| = |E|/Z$ from the impedance relation (Lecture 2).

② ⇒ $\vec{E}_\parallel + \vec{E}_\parallel'' = \vec{E}_\parallel'$ *E is along $\hat{y}$, i.e. **entirely** tangential → no cos*

$$E_0 + E_0'' = E_0' \qquad \text{...(I)}$$

④ ⇒ $\vec{H}_\parallel + \vec{H}_\parallel'' = \vec{H}_\parallel'$ *H lies **in** the plane → tangential part = H cos*

$$\frac{E_0}{Z}\cos i - \frac{E_0''}{Z}\cos i = \frac{E_0'}{Z'}\cos r \qquad \text{...(II)}$$

The minus sign: $\mathbf{H} \propto \hat{k} \times \mathbf{E}$, and the incident and reflected waves travel opposite ways in z, so their tangential ($\hat{x}$) parts point opposite ways — exactly the same minus that appears in the professor's p-polarization equation $E_0 \cos i - E_0'' \cos i = E_0' \cos r$.

*Compare with his p case: there **H** was ⊥ the plane (no cos) and **E** was in it (with cos). For s the two swap over. (I) and (II) are Jackson's Eq. (7.38).*

Step 4 — Solve the two equations

Use (I) to remove E_0'', then read off E_0'.

From (I): $E_0'' = E_0' - E_0$. Put into (II):

$$\frac{\cos i}{Z}(2E_0 - E_0') = \frac{\cos r}{Z'}E_0' \Rightarrow \frac{2E_0\cos i}{Z} = E_0'\frac{Z'\cos i + Z\cos r}{ZZ'}$$

STEP 4 — halfway, not Eq. (7.39) yet

$$\frac{E_0'}{E_0} = \frac{2Z'\cos i}{Z'\cos i + Z\cos r}, \quad \frac{E_0''}{E_0} = \frac{E_0'}{E_0} - 1 = \frac{Z'\cos i - Z\cos r}{Z'\cos i + Z\cos r}$$

This matches the professor's Lecture 4 slide exactly — but it still contains Z, Z′ and cos r. Eq. (7.39) uses n, n′ and i only, so one more step is needed.

Step 5 — Rewrite it the way Jackson does

Two substitutions: replace Z by n, then remove r using Snell.

(a) $n = c\sqrt{\mu\varepsilon} \Rightarrow Z = \sqrt{\mu/\varepsilon} = \frac{\mu}{\sqrt{\mu\varepsilon}} = \frac{\mu c}{n}$, $Z' = \frac{\mu' c}{n'}$

Both come straight off one slide (Lec 2, plane waves in a nonconducting medium): *$v_p = 1/\sqrt{(\mu\varepsilon)}$ and $n = c/v_p$ give $\sqrt{(\mu\varepsilon)} = n/c$, and $Z = \sqrt{(\mu/\varepsilon)}$. The one trick is to multiply inside the root by μ: $\sqrt{(\mu/\varepsilon)} = \sqrt{(\mu^2/\mu\varepsilon)} = \mu/\sqrt{(\mu\varepsilon)} = \mu/(n/c) = \mu c/n$.*

$$\frac{E_0'}{E_0} = \frac{2(\mu' c/n')\cos i}{(\mu' c/n')\cos i + (\mu c/n)\cos r} = \frac{2n\cos i}{n\cos i + (\mu/\mu')n'\cos r} \qquad \times nn'/(\mu' c)$$

(b) Snell $n\sin i = n'\sin r \Rightarrow n'\cos r = \sqrt{n'^2 - n^2\sin^2 i}$

FINAL ANSWER = JACKSON (7.39)

$$\frac{E_0'}{E_0} = \frac{2n\cos i}{n\cos i + \frac{\mu}{\mu'}\sqrt{n'^2 - n^2\sin^2 i}}$$

$$\frac{E_0''}{E_0} = \frac{n\cos i - \frac{\mu}{\mu'}\sqrt{n'^2 - n^2\sin^2 i}}{n\cos i + \frac{\mu}{\mu'}\sqrt{n'^2 - n^2\sin^2 i}}$$

■ END OF REQUIRED SOLUTION — everything below is optional ■

OPTIONAL **Quick checks.** No interface ($n = n'$): $Z = Z'$, $r = i \Rightarrow E_0''/E_0 = 0$, $E_0'/E_0 = 1$. ✓ Energy: $R + T = 1$, using $(a-b)^2 + 4ab = (a+b)^2$ with $a = Z'\cos i$, $b = Z\cos r$. ✓

Problem 2 — Brewster's angle

> Show that for **p polarization** (E parallel to the plane of incidence) there is an angle at which nothing is reflected, and find it. Take μ = μ′. — Jackson Eq. (7.43).

ANSWER

$$i_B = \tan^{-1}\frac{n'}{n}$$

Only p polarization has such an angle. Air → glass: 56.3°; glass → air: 33.7°.

Step 0 — Where does this start?

*You do **not** redo Problem 1. The professor already derived the p-polarization coefficient in class by exactly the same four-boundary-condition route; you start from his result.*

$$\frac{E_0''}{E_0} = \frac{Z\cos i - Z'\cos r}{Z\cos i + Z'\cos r}$$ Lec 4, slides 3 & 13 — *Z and Z′ swapped vs. s*

Compare with the s result on p. 4: the only difference is which impedance sits with cos i. That swap is the entire reason Brewster's angle exists for p and not for s.

Step 1 — When is the reflected amplitude zero?

A fraction is zero when its numerator is zero.

$E_0'' = 0 \Leftrightarrow Z\cos i = Z'\cos r$

Step 2 — Write it in terms of n

Put Z = μc/n (same bridge as Problem 1, Step 5). With μ = μ′ the μc cancels.

$$\frac{\mu c}{n}\cos i = \frac{\mu c}{n'}\cos r \Rightarrow \boxed{n'\cos i = n\cos r} \quad ...(A)$$

and Snell always holds: $\boxed{n\sin i = n'\sin r}$ ...(B)

Step 3 — Combine (A) and (B)

Multiply them together — the n's cancel and a double angle appears.

$n\,n'\sin i\cos i = n\,n'\sin r\cos r \;\Rightarrow\; \sin i\cos i = \sin r\cos r \;\Rightarrow\; \sin 2i = \sin 2r$

Two possibilities: $2i = 2r$ (means $n = n'$, no interface — reject), or $2i = 180° - 2r$:

$$i_B + r = 90°$$

Step 4 — Put that back into Snell

$n\sin i_B = n'\sin(90° - i_B) = n'\cos i_B \;\Rightarrow\; \tan i_B = n'/n$

FINAL ANSWER = JACKSON (7.43)

$$i_B = \tan^{-1}\frac{n'}{n} \qquad \text{(equivalently } i_B + r = 90°\text{)}$$

■ END OF REQUIRED SOLUTION — everything below is optional ■

OPTIONAL **Same answer without the double-angle trick.** From (A) $\cos r = (n'/n)\cos i$ and from (B) $\sin r = (n/n')\sin i$. Then $\sin^2 r + \cos^2 r = 1$ gives $(n^2/n'^2)\sin^2 i + (n'^2/n^2)\cos^2 i = 1$, hence $\sin^2 i_B = n'^2/(n^2+n'^2)$ and $\cos^2 i_B = n^2/(n^2+n'^2)$, so $\tan i_B = n'/n$. ✓

EXTRA — none of this page is required for the answer

Why s polarization has no Brewster angle

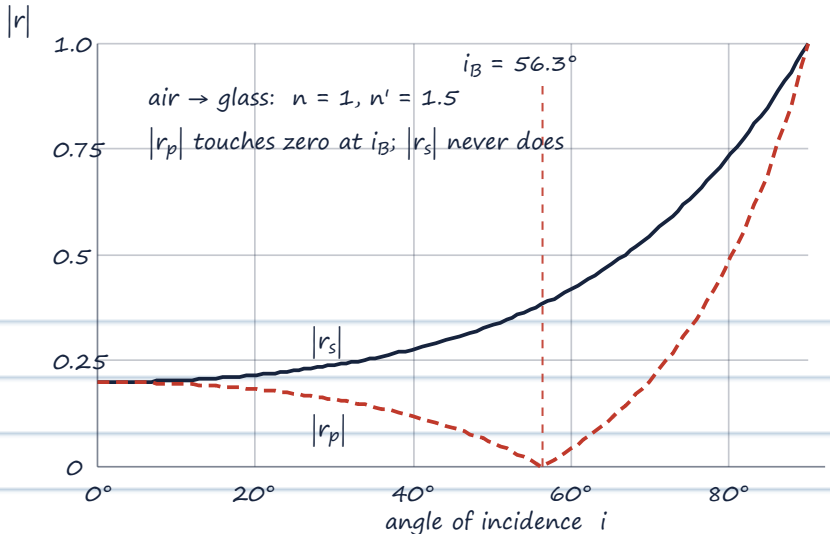
From Problem 1, $E_0'' = 0$ would need $Z'\cos i = Z\cos r$, i.e. (with $\mu = \mu'$) $n\cos i = n'\cos r$.

Combine with Snell $n\sin i = n'\sin r$ — **square both and add**: $n^2 = n'^2$ — only if there is no interface at all.

So r_s never vanishes, at any angle. Light reflected at i_B is therefore 100% s polarized — that is how polarizing sunglasses and camera filters remove glare.

Physical picture

*$i_B + r = 90°$ means the reflected direction is perpendicular to the refracted one. For p polarization the transmitted **E′** is perpendicular to **k′**, so it points exactly along the reflected direction — and **a dipole radiates nothing along its own axis.***



Same shape as the left panel of the Lecture 4 "Brewster's angle" slide.

OPTIONAL **Number check.** Air → glass: $i_B = \tan^{-1}1.5 = 56.3°$; Jackson writes "≈ 56°". ✓
Glass → air: 33.7°, below the critical angle 41.8°. ✓